DURHAM
COLLEGE

COSC 1100

Class Exercise

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Class Exercise

Numeric and String Data: A Math-Based Solution

Part 1 of 2

Numeric and String Data:
A Math-Based Solution

Goal of Part 1:

Experience designing a program that requires a mathematical solution, while reinforcing the importance of our planning process.

Requirements

- Did you come prepared?
  - Numeric and String Data videos, hopefully done or almost done!

- Collaboration.
  - You will be encouraged to work together with 1-2 colleagues during this exercise. (i.e. groups of 2-3)
  - Students currently outside of Canada are encouraged to communicate and collaborate via MS Teams.
  - The group will share their planning ideas at the end of this class, which will contribute to your grade.

A Simple Programming Approach Yet Again (Steps 1 – 5)

1. Clearly define the result (**OUTPUT**)

2. Determine what data is needed, and how to get it (**INPUT**)

3. List the steps needed in between (**PROCESS**)

4. Write your **PLAN** as pseudo-code and/or a flowchart

5. Test your plan for logical errors (**DESK-CHECK**)

How to Complete Part 1 of This Class Exercise

- Read the following problem statement, *for detailed comprehension*.

1. Analyze the problem requirements, compile a reasonable plan to create a solution, and check it thoroughly.
   In other words, finish steps 1 - 5 of our planning approach.
2. Be prepared to discuss your notes in class.
3. Make a discussion post including your notes on DC Connect (**COSC 1100** -> **Activities** -> **Discussion** -> **Class Exercises**), including all group members' names.

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The Problem

Durham College likes to promote its applied research centres, with promotional items such as AI Hub-branded rubber ducks. As somewhat of a publicity stunt, someone has floated the idea of completely covering one of the two reservoirs at Durham College's Oshawa Campus with these ducks.

Can you estimate the size of these reservoirs, and then use a program to determine how many rubber ducks would be required to cover it almost completely?

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Class Exercise Part 1 Discussion

Consider the notes presented and things other students have said.

- Were there any groups who had significantly different input than your group?
- Were there any groups who had significantly different output than your group?
- If any of these were different, *why* do you think they were different?

Class Exercise Part 1 Summary

- You should have a finished plan that can tell how many ducks fit on a given size of body of water, somehow!
  (Keeping in mind these slides are prepared in advance, it's fully possible that your approach was very different, but it should be something along those lines…)

- Be ready to discuss the next few steps – including coding this – very shortly.

- Do you feel *ready* to code this?

COSC 1100
Class Exercise

Numeric and String Data: A Math-Based Solution

Part 2 of 2

Numeric and String Data:
A Math-Based Solution

Goal of Part 2:

Experience working toward a complete, working program from a plan, with emphasis on mathematics and string formatting.

Requirements

- Did you come prepared?
  - Numeric and String Data videos, ideally all done!

- Collaboration.
  - You may continue to work together with 1-2 colleagues during this exercise. (i.e. groups of 2-3)
  - Students currently outside of Canada are encouraged to communicate and collaborate via MS Teams.
  - Each group will submit their finished application for marks.

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The Problem (Review)

Durham College likes to promote its applied research centres, with promotional items such as AI Hub-branded rubber ducks. As somewhat of a publicity stunt, someone has floated the idea of completely covering one of the two reservoirs at Durham College's Oshawa Campus with these ducks.

Can you estimate the size of these reservoirs, and then use a program to determine how many rubber ducks would be required to cover it almost completely?

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Plan to Program!

- With consideration to the plan from yesterday, code the application using Python!

1. Code your solution to the given problem based on the plan provided or your own planning notes.
2. Chat, ask lots of questions, and don't be afraid of mistakes.
3. Either have it checked during class,
   or submit your group's finished Python code file to DC Connect
   (**COSC 1100** -> **Activities** -> **Assignments** -> **Class Exercise: Numeric and String Data**), including all group members' names.

This is meant to be done during class time; submissions received after our class time ends may be considered late.

Class Exercise Part 2 Discussion

Consider the process of creating today's Python application.

- Suppose this was a completely real and legitimate problem you were presented with.
  Would you actually use a program to solve it?
  What other option(s) might you consider, and what are the pros and cons of that?

- Did the desk check from Step 5 of the plan bring you any benefit? Why or why not?

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Class Exercise Part 2 Summary

- You’ve created an application that automates some relatively complex math, and formats the output nicely!

- Get used to finding and using mathematical formulas in your program, and paying attention to things like units and decimal places!